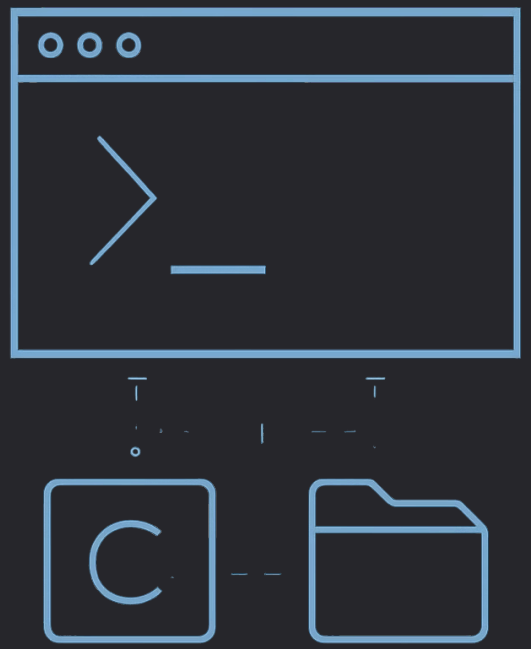


PROJECT TITLE

Interactive Linux Automation Shell

A Custom Command-Line Interface for OS Concepts

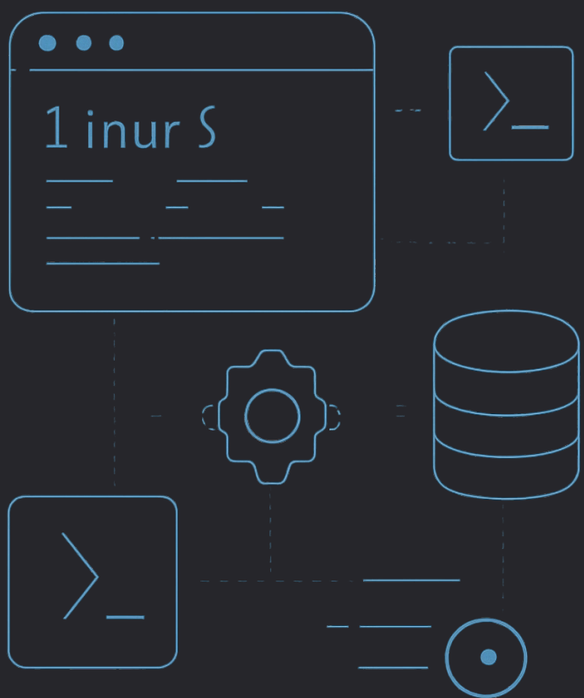
Team Information

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Department: CSE SEC-7

Guided By: P kalpana

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OVERVIEW

What is Interactive Linux Automation Shell?

A custom command-line interface developed in C that enables users to execute Linux commands and automate complex tasks — providing hands-on experience in core Operating System concepts.

Process Management

Understanding fork, exec, and process lifecycle

System Calls

Direct interaction with the Linux kernel

Shell Programming

Building a functional shell interpreter

Command Execution

Parsing and executing user commands efficiently



User Input | Command Parsing | Validation | Execution | Output

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Project Objectives



Custom Linux Shell

Develop a custom Linux shell in C



System Calls

Execute Linux commands using system calls



Process Management

Implement fork(), exec(), and wait() system calls



Pipes & Redirection

Support pipes and I/O redirection



Background Processes

Manage background processes and signal handling



OS Knowledge

Gain practical understanding of Operating System concepts

"We also plan to include custom commands that automate repetitive tasks and optionally display the actual Linux commands, making the shell easier for beginners to learn."

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Technology Stack



Programming Language: C

Core language for system-level programming.



Operating System: Linux Ubuntu

Robust open-source OS for development.



Compiler: GCC

GNU Compiler Collection for C.



Development Tools: VS Code / Vim

Powerful code editors for efficient development.



Debugger: GDB

GNU Debugger for C programs.



Build Tool: Makefile

Automates compilation and linking processes.



Libraries: POSIX System Calls

Standard API for OS interaction.

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Expected Outcomes

Functional Shell

Develop a fully functional Interactive Linux Automation Shell

System Calls

Execute Linux commands efficiently using system calls

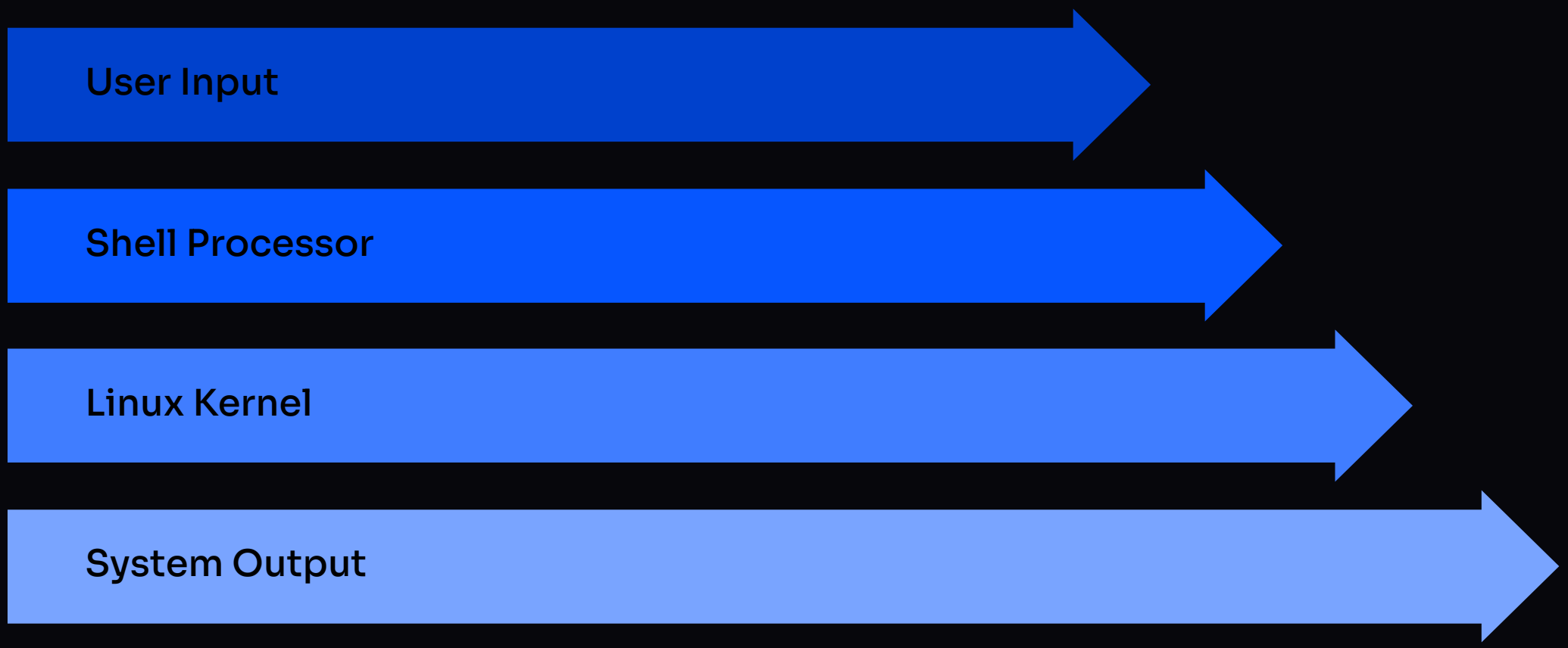
Advanced Features

Implement process management, pipelines, and I/O redirection

OS Knowledge

Gain practical knowledge of Operating System concepts

Workflow



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Project Modules

A hub-and-spoke architecture diagram showing the core components of the Interactive Linux Automation Shell.

1 **Interactive Linux Automation Shell**

Command Executor

fork(), exec(), wait()

Pipe & I/O Redirection

Signal Handler

Command Parser

Built-in Commands

cd, pwd, exit, history

Job Control

Foreground & Background Processes

Shell Script Executor

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Project Implementation Timeline

Week 1

Requirement Analysis & Project Design

Week 2

Command Parser & Basic Shell

Week 3

Process Management (fork, exec, wait)

Week 4

Pipes & I/O Redirection

Week 5

Background Jobs & Signal Handling

Week 6

Shell Scripting, Testing, Debugging & Final Integration



This timeline outlines the key development phases, from initial design to final integration, ensuring a structured and efficient project delivery.

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